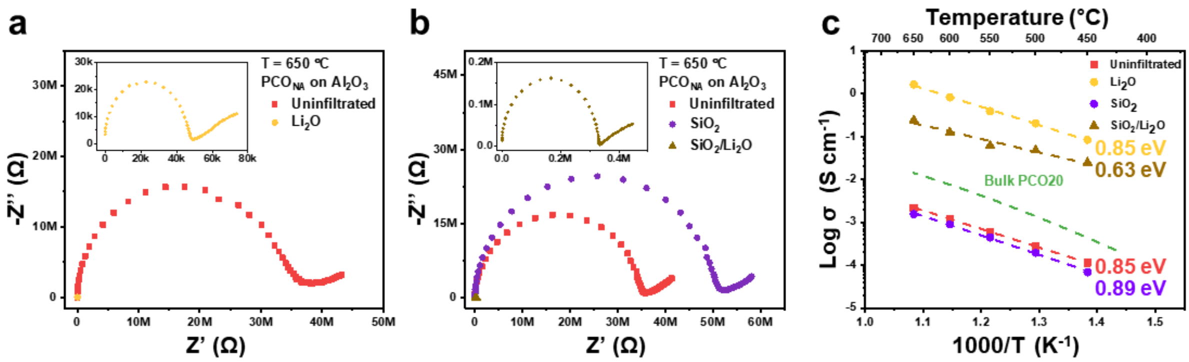

Fig. 2 Influence of infiltrated binary oxide acidity on PCO$_{NA}$ electronic properties.
a Impedance spectra of pristine PCO$_{NA}$ followed by Li-infiltration at 650 °C (see inset for reduced resistance). **b** Impedance spectra of pristine PCO$_{NA}$ followed by Si-infiltration and serial Li-infiltration for recovering conductance (see inset for reduced resistance) at 650 °C. Al_2O_3 used as substrate in both cases. **c** Arrhenius plots of in-plane conductivity of pristine PCO$_{NA}$ followed by Li-, Si- and serial Li-infiltrations along with their respective activation energies, respectively. The green dashed line indicates the conductivity behavior of a bulk PCO20 film calculated based on defect chemistry modelling.

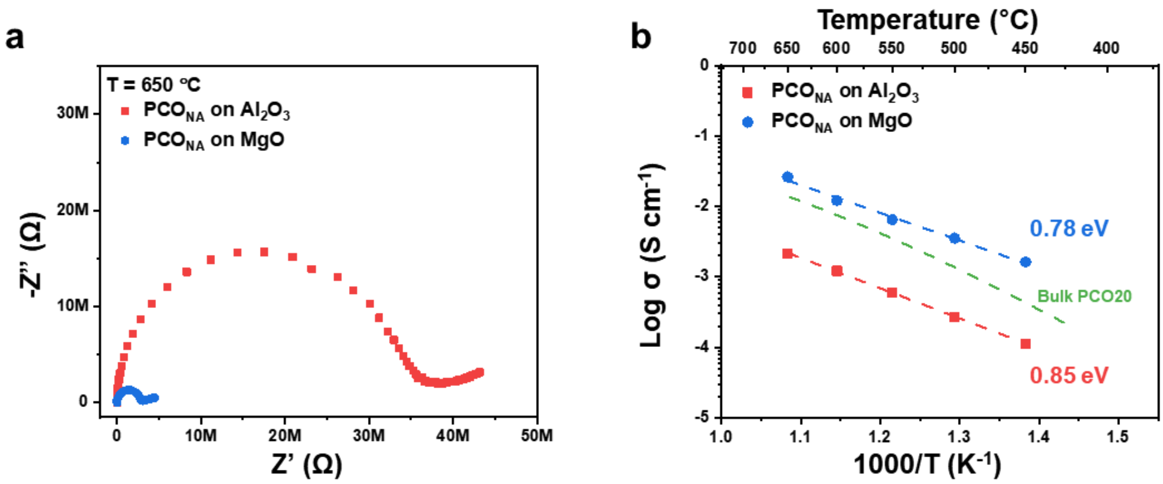

Fig. 3 Influence of insulating substrate acidity on PCO$_{NA}$ electronic properties.

a Comparison of impedance spectra of PCO$_{NA}$ fabricated on Al_2O_3 and MgO substrates measured at 650 °C. **b** Arrhenius plots of in-plane conductivity of PCO$_{NA}$ fabricated on Al_2O_3 and MgO substrates along with their respective activation energies. The green dashed line indicates the conductivity behavior of a bulk PCO20 film calculated based on defect chemistry modelling.

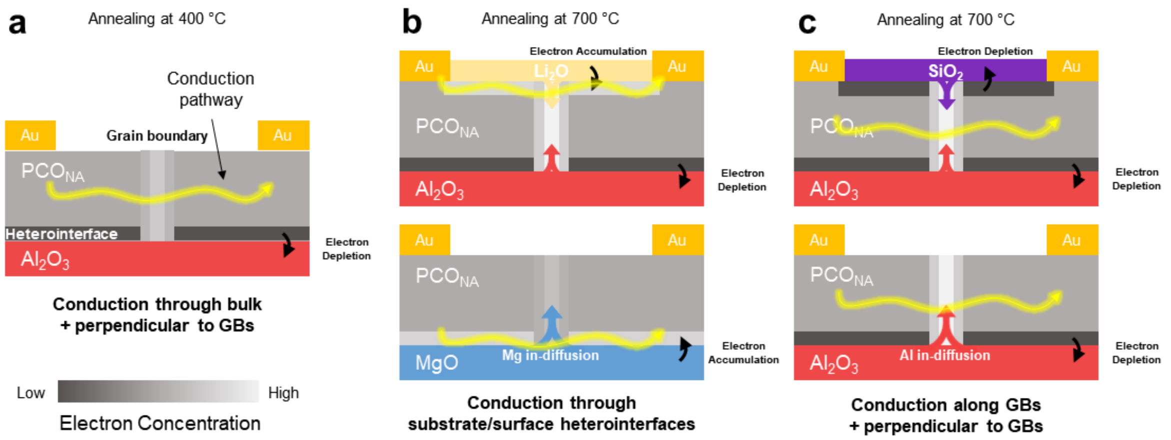

Fig. 4 Schematic illustration of heterointerface acidity-mediated space charge potentials at heterointerfaces and GBs of PCO_{NA}, and impact of electron redistribution

a PCO_{NA} annealed at 400 °C. Conduction mainly occurs through bulk (R_1) and perpendicular to GBs (R_4). Substrate heterointerface shown in black corresponds to electron depletion due to the relative acidity of Al_2O_3. Slightly brighter GB is attributed to a built-in space charge potential (positive core charge) leading to some electron accumulation. For high-temperature annealing, cation diffusion into GBs must be considered. Diffusion of cations up to GBs leads to GB electron accumulation/depletion, described in white/gray (see **b** and **c**). **b** Li-infiltrated PCO_{NA} (upper image) and PCO_{NA} fabricated on MgO substrate (lower image) annealed at 700 °C. Conduction mainly occurs through substrate (R_2) (upper image) or surface (R_5) (lower image) heterointerfaces. Substrate/surface heterointerface shown in light grey corresponds to electron accumulation due to the relative basicity of Li_2O and MgO. **c** Si-infiltrated PCO_{NA} (upper image) and PCO_{NA} fabricated on Al_2O_3 substrate (lower image) annealed at 700 °C. Conduction mainly occurs along GBs (R_3) and perpendicular to GBs (R_4). Substrate/surface heterointerface shown in black corresponds to electron depletion due to the relative acidity of SiO_2 and Al_2O_3. The scale of electron concentration is represented through the color bar at the bottom left.